

232.2.4971 - Quantum Computation

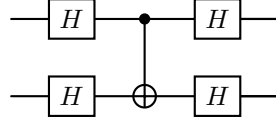
Homework Assignment 02

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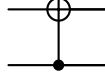
Due: 27/05/2026 - extended

Question 1 - Multiple qubits and quantum circuits

For each state $|x_0, x_1\rangle$ for $x_i \in \{0, 1\}$, apply the following circuit:



Repeat the calculation with the following circuit:



Conclude that these two unitary transformations are equal, and that the intuition that the C-NOT gate does not change the control qubit is false, when we work in the $|+\rangle, |-\rangle$ basis.

Solution

Let us define:

$$\begin{aligned} H_2 &= H \otimes H \\ C &= CNOT \\ \tilde{C} &= CNOT_{2 \rightarrow 1} \end{aligned}$$

Where H is Hadamard operator, $CNOT$ is controlled-not where first qubit is control and second qubit is target. $CNOT_{2 \rightarrow 1}$ is controlled-not where second qubit is control and first is the target. Applying the first circuit is equivalent to performing the H_2CH_2 operator:

For state $|00\rangle$:

$$\begin{aligned} H_2CH_2|00\rangle &= H_2C|++\rangle \\ &= \frac{1}{2}H_2C(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \\ &= \frac{1}{2}H_2(|00\rangle + |01\rangle + |11\rangle + |10\rangle) \\ &= H_2|++\rangle \\ &= |00\rangle \end{aligned}$$

For state $|01\rangle$:

$$\begin{aligned} H_2CH_2|01\rangle &= H_2C|+-\rangle \\ &= \frac{1}{2}H_2C(|00\rangle - |01\rangle + |10\rangle - |11\rangle) \\ &= \frac{1}{2}H_2(|00\rangle - |01\rangle + |11\rangle - |10\rangle) \\ &= \frac{1}{2}H_2(|0\rangle(|0\rangle - |1\rangle) + |1\rangle(|1\rangle - |0\rangle)) \\ &= \frac{1}{2}H_2(|0\rangle - |1\rangle)(|0\rangle - |1\rangle) \\ &= H_2|--\rangle \\ &= |11\rangle \end{aligned}$$

For state $|10\rangle$:

$$\begin{aligned}
H_2CH_2|10\rangle &= H_2C| -+\rangle \\
&= \frac{1}{2}H_2C(|00\rangle + |01\rangle - |10\rangle - |11\rangle) \\
&= \frac{1}{2}H_2(|00\rangle + |01\rangle - |11\rangle - |10\rangle) \\
&= H_2| -+\rangle \\
&= |10\rangle
\end{aligned}$$

For state $|11\rangle$:

$$\begin{aligned}
H_2CH_2|01\rangle &= H_2C|+-\rangle \\
&= \frac{1}{2}H_2C(|00\rangle - |01\rangle - |10\rangle + |11\rangle) \\
&= \frac{1}{2}H_2(|00\rangle - |01\rangle - |11\rangle + |10\rangle) \\
&= \frac{1}{2}H_2(|0\rangle(|0\rangle - |1\rangle) - |1\rangle(|1\rangle - |0\rangle)) \\
&= \frac{1}{2}H_2(|0\rangle + |1\rangle(|0\rangle - |1\rangle)) \\
&= H_2|+-\rangle \\
&= |01\rangle
\end{aligned}$$

The second circuit is equivalent to performing the $\tilde{C}$ operator, this by definition gives us the following behavior:

$$\begin{aligned}
\tilde{C}|00\rangle &= |00\rangle \\
\tilde{C}|01\rangle &= |11\rangle \\
\tilde{C}|10\rangle &= |10\rangle \\
\tilde{C}|11\rangle &= |01\rangle
\end{aligned}$$

Comparing the results of the two circuits we observe the same behavior for all 4 base states, therefore concluding they are equal. Furthermore, observing (for example) the behavior of the first circuit on the state $|01\rangle$ we notice that the following is true:

$$C|+-\rangle = |--\rangle$$

This demonstrates a case in which, when operating in the $\{+, -\}$ basis, C-NOT changes the control (first) qubit

Question 2 - Do we really need complex numbers

Prove that we can assume w.l.o.g. that all amplitudes in a quantum computation are real numbers!

Guidance: Show that any quantum circuit on n qubits that uses t two-qubit gates with complex coefficients can be simulated exactly by a quantum circuit on $n+1$ qubits, that uses at most t three-qubit gates with real coefficients. Show that the probability of acceptance remains the same in your construction. Hint: Remember that we assume that the input for a quantum circuit is in the standard basis (so, in the base of our induction, we know that the state has only real amplitudes). Use the extra qubit to decide whether the original state is in the Imaginary or Real part, and adapt each gate to a gate with one more qubit, such that the new gate works in an equivalent way to the old one. More formally, suppose that at some step of the original circuit the state at that step is $\sum_x (a_x + ib_x)|x\rangle$, the state at that step in the new circuit should be $\sum_x a_x|0x\rangle + b_x|1x\rangle$.

Solution

Let M be a map that acts on states and operators as follows:

$$M(|c_k\rangle) = \sum_x (a_x + ib_x) |x\rangle = |r_k\rangle = \sum_x a_x |0x\rangle + b_x |1x\rangle$$

$$M(A + iB) = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}$$

To prove that we can assume w.l.o.g that all amplitudes in a quantum computation are real numbers we need to show that for any given circuit c with complex amplitudes there exists an equivalent circuit r with real amplitudes - meaning that r and c have the same probabilities

Induction on the number of gates:

Base case ($k = 0$): The input state is a computational basis state, hence has real amplitudes. Therefore, by the quantum circuit definition, the circuit's input is in the standard basis and is classical - therefore the input's amplitudes are real hence the inputs of c and r are trivially equivalent (i.e. $c_0 = r_0$).

Induction hypothesis: Assume that for some $|k\rangle$ in circuit c , $M(|k\rangle)$ in circuit r gives the same probabilities

Induction step: Let the next gate be U_{k+1} , then:

$$\begin{aligned} |c_{k+1}\rangle &= U_{k+1} |c_k\rangle \\ &= (A + iB) \left(\sum_x a_x + ib_x \right) |x\rangle \\ &= \left(\sum_x (A_x a_x - B_x b_x) + i(A_x b_x + B_x a_x) \right) |x\rangle \\ |r_{k+1}\rangle &= G_{k+1}^r |r_k\rangle \\ &= G_{k+1}^r \sum_x (a_x |0x\rangle + b_x |1x\rangle) \\ &= \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \left(\sum_x a_x |0x\rangle + b_x |1x\rangle \right) \\ &= \sum_x ((A_x a_x - B_x b_x) |0x\rangle + (A_x b_x + B_x a_x) |1x\rangle) \end{aligned}$$

From above, we get $M(|c_{k+1}\rangle) = |r_{k+1}\rangle$. The probability to get state $|j\rangle$ in circuit c is:

$$\langle k | c_{k+1} \rangle = |(A_x a_x - B_x b_x) + i(A_x b_x + B_x a_x)|^2 = (A_x a_x - B_x b_x)^2 + (A_x b_x + B_x a_x)^2$$

The probability to get state $|j\rangle$ in circuit r is:

$$\langle k | r_{k+1} \rangle = (A_x a_x - B_x b_x)^2 + (A_x b_x + B_x a_x)^2$$

Which results in the same probability. QED.

Question 3 - Reconstructing a matrix

Let A be a binary $n \times n$ matrix. For $x, y \in \{0, 1\}^n$, let:

$$f(x, y) = x^T A y \bmod 2 = \sum_{i,j} (A_{ij} x_i y_j) \bmod 2.$$

Convince yourself (there is no need to submit this argument) that there exists a simple classical algorithm that reconstructs A using n^2 queries to f , and that there is no algorithm that can do that with fewer classical queries.

Find a quantum algorithm that reconstructs the matrix A using $O(n)$ queries to the corresponding quantum phase oracle. Recall that the quantum phase oracle for f acts as follows:

$$\forall x, y \in \{0, 1\}^n, V_f |x, y\rangle = (-1)^{\sum_{i,j} A_{ij} x_i y_j \bmod 2} |x, y\rangle$$

Solution

The main idea is to query each column of A with Bernstein–Vazirani algorithm:

1. Prepare the state $|0\rangle^{\times n} |e_j\rangle$ where e_j is the j -th standard basis vector.
2. Apply Hadamard on the n $|0\rangle$ qubits to get: $\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle |e_j\rangle$
3. Apply the oracle V_f to get: $\frac{1}{\sqrt{2^n}} \sum_x (-1)^{s_j \cdot x} |x\rangle |e_j\rangle$ - the e_j extracts the j -th column which is queried by the oracle
4. Apply Hadamard again on the first n qubits and measure - the answer is column j .
5. Repeat the algorithm for every j - this is repeated n times, each time receiving another column, and therefore has complexity of $O(n)$

question-4—grover's-algorithm-with-an-unknown-number-of-marked-items

You are given access to an oracle with K marked items, and K is unknown. Design an algorithm that finds a marked item using $O(\sqrt{N/K})$ queries, with probability more than $\frac{1}{2}$. Guidance:

1. Show that when K is known, there is an algorithm that finds the marked element with probability close to 1, with running time $O(\sqrt{N/K})$.
2. Show that if K lies between 2^j and 2^{j+1} for a known j , there is an algorithm that finds a marked element with probability at least $\frac{1}{2}$, with the same running time.
3. Iterate over the possible values of j in a certain order, which gives a total running time of $O(\sqrt{N/K})$ with probability $\frac{1}{2}$.

Solution

K is known:

If K is known then we can mark $J = \{i \mid x_i = 1\}$ the set of marked elements. Similar to Grover's algorithm we define:

$$|\alpha\rangle = 1/\sqrt{N-K} \sum_{i \in 0,1^n \setminus J} |i\rangle$$

$$|\beta\rangle = 1/\sqrt{K} \sum_{i \in J} |i\rangle$$

Applying Grover's algorithm with $\lfloor \frac{\pi\sqrt{N/K}}{4} - 1/2 \rfloor$ repetitions yields the correct solution (finding a marked character) with optimum probability - close to 1 (the calculation is similar to that in the original algorithm where $K = 1$).

K is between $2^j, 2^{j+1}$:

In this case we choose to iterate Grover's algorithm for $K = 2^j$ and the success probability is:

$$\begin{aligned} P(\text{hit}) &= \sin((2t+1)\theta) \\ &= \sin\left(2\left(\frac{\pi}{4}\sqrt{N/K} - 1/2\right) + 1\right)\theta \\ &= \sin\left(\left(\frac{\pi}{2}\sqrt{N/2^j}\right)\sqrt{K/N}\right) \\ &= \sin\left(\frac{\pi}{2}\sqrt{K/2^j}\right) \end{aligned}$$

But since $2^j \leq K \leq 2^{j+1}$ then the probability is $P(hit) \geq 1/2$ since we are off by at most a factor of 2 and therefore the angle is in the range of $[\pi/4, 3\pi/4]$.

K is unknown:

In this case we divide the space of possible K into $\log N$ segments. In each segment K is bounded by $2^j \leq K \leq 2^{j+1}$, and we run the algorithm from step 2 on each segment until we find a match.

The running time for stage j is $O(\sqrt{N/2^j})$. Since the cost decreases as j increases, we start guessing from $j = \log N$ and go down. Suppose the true value satisfies $2^{j^*} \leq K < 2^{j^*+1}$ and the total work up to stage j^* is:

$$O\left(\sum_{j=0}^{j^*} \sqrt{N/2^j}\right) = O\left(\sqrt{N} \sum_{j=0}^{j^*} \sqrt{1/2^j}\right) = O\left(\sqrt{N/2^{j^*}}\right)$$

The last transition uses the fact that the sum is descending geometric sum and is bounded, and since K Now since $2^{j^*} \leq K$ we get a final complexity of $O(\sqrt{N/K})$